SOBA Observable Public-Evidence Ballot Audits

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Abstract

Stylish Observable Ballot-level Audit (SOBA) evidence makes the comparison between cast vote records and ballot-card interpretations publicly observable, usually by publishing commitments and later opening the sampled records without revealing voter identity. RCOUNT implements the first boundary of that idea as a record-shape and observable-opening contract. A SOBA audit run must use the versioned method id `soba-observable-ballot-audit-v1`, carry observable ballot linkage assertions, name at least one sampled opening, and link every sample step to a privacy-safe inclusion proof. The verifier checks that each referenced opening exists and that the proof does not publish candidate choices or voter identity. Statistical comparison-risk replay is deliberately outside this slice; the audit crate records SOBA linkage as a documented boundary transcript.

1 Introduction

Ballot-level audits are most useful to the public when the comparison can be checked without trusting the auditor's private software or notes. In a SOBA-style audit, an observer should be able to see that a sampled cast vote record has a corresponding ballot-card interpretation, and that the comparison was made against a committed public record rather than a record chosen after the fact.

RCOUNT implements the first public-evidence layer of that idea. The current contract is not a full SOBA statistical audit and does not recompute a comparison risk measure. It asks a narrower question: does the audit record declare an observable ballot-linkage method, and does each sampled unit name a privacy-safe opening that is actually present in the package?

This matters because observability and secrecy pull in opposite directions. The record must be open enough for a public observer to check the CVR-to-ballot comparison path, but not so open that it publishes a vote receipt or a voter identifier. The implemented slice therefore treats SOBA as a boundary contract for record shape, opening linkage, and privacy-safe proof contents.

2 Observable Contract

The implemented record is an audit algorithm run whose method id is `soba-observable-ballot-audit-v1`. It carries observable ballot linkage assertions and sample steps that point to public openings.

In the synthetic fixture, the opening is represented by an inclusion proof id; the sample step names that proof as its `sample_unit_id`.

Field	Meaning
`method_id`	Versioned SOBA boundary method. Current value: `soba-observable-ballot-audit-v1`.
`assertions`	Non-empty list of `ObservableBallotLinkage` assertions.
`assorter_id`	Fixture vocabulary for the observable opening, currently `soba-commitment-opening-v1`.
`sample_steps`	Non-empty ordered list of sampled observable records.
`sample_unit_id`	Proof id of the public opening or commitment-opening record.
`source_refs`	Source evidence for the SOBA run and the sampled opening.
`inclusion_proofs`	Package-level proof records, keyed by `proof_id`.
`candidate_selections`	Must be empty on the linked proof; the opening must not publish choices.
`voter_id`	Must be absent on the linked proof; the opening must not identify a voter.

The contract is therefore about observable comparison fields and privacy-safe openings, not about performing the comparison statistic. It verifies that the public record contains the declared opening and that the opening keeps the same receipt-safety boundary as the V.4 inclusion-proof vocabulary.

3 Implemented Surface

The core verifier calls `verify_soba_observable_ballot_design` for every audit algorithm run. The method id is exported as `SOBA_OBSERVABLE_BALLOT_AUDIT_METHOD_ID` and has the value `soba-observable-ballot-audit-v1`.

The function enforces the boundary exactly. If a non-SOBA run contains any `ObservableBallotLinkag` assertion, it is rejected as an invalid observable-ballot design. If a SOBA run has no assertions, no sample steps, or any assertion whose kind is not `ObservableBallotLinkage`, it is also rejected. The verifier then indexes `package.inclusion_proofs` by `proof_id`. For each sample step, `sample_unit_id` must name an existing proof. A missing proof raises `MissingObservableBallotOpening` with the run id, step index, and proof id. Finally, each linked proof must have empty `candidate_selections` and no `voter_id`; otherwise the run is rejected as an invalid observable-ballot design.

The audit crate exposes the same method id through `replay_audit_algorithm_statistics`. That surface returns a boundary transcript saying that SOBA observable-ballot linkage is recorded but not comparison-risk replay. This is intentional: the implemented code verifies the observable record and opening linkage, not full ballot-level comparison audit statistics.

Check	Failure caught
audit_algorithm_transcript	SOBA run has no assertions, no sample steps, non-observable assertions, invalid assorter bounds, invalid sample values, or invalid p-values.
verify_soba_observable_ballot_design	Observable linkage appears under the wrong method id, the named opening is missing, or the linked proof publishes choices or voter identity.
replay_audit_algorithm_statistics	SOBA is reported as a documented boundary rather than as comparison-risk replay.
source_hash_match	Source evidence bytes still match the declared source index before the audit record is trusted.

4 Fixtures

The positive boundary fixture is `synthetic_soba_observable_ballot_boundary_package`. It starts from the privacy-safe inclusion-proof package and adds one audit algorithm run: `audit-run:soba-observable-ballot-boundary`. The run uses `soba-observable-ballot-audit-v1`, sets `sampling_mode` to `BoundaryOnly`, declares one `ObservableBallotLinkage` assertion with `soba-commitment-opening-v1`, and adds one sample step whose `sample_unit_id` is `proof:accepted-token-001`. That proof exists in the package and has no candidate selections and no voter id, so the package verifies and records an `audit_algorithm_transcript` pass.

The negative fixture is `synthetic_missing_soba_opening_package`. It starts from the same boundary package but changes the sample step to `proof:missing-token`. The verifier can still parse the run, assertion, and step, but it cannot find the named opening in `inclusion_proofs`. The package is rejected with `MissingObservableBallotOpening`.

Fixture	Boundary exercised
synthetic_soba_observable_ballot_boundary_package	Present opening, observable linkage assertion, privacy-safe proof contents.
synthetic_missing_soba_opening_package	Sample step references an opening that is not in the package.

The corresponding tests assert both sides: the boundary package verifies and the linked proof remains choice-free, while the missing-opening package fails with the specific missing-opening error.

5 Boundaries

This slice is a record-shape and observable-opening contract. It verifies that a SOBA audit run declares the right method id, uses observable ballot linkage assertions, includes sample steps, links every sampled unit to a package proof, and keeps those proofs from publishing candidate selections or voter identity. It does not verify a cryptographic commitment scheme, compare CVRs to ballot interpretations, or compute a risk-limiting comparison statistic.

That boundary matches the implementation plan. RCOUNT can verify commitments and transcript consistency only to the extent those fields exist in the public package and are linked by stable ids. The current fixture lands the opening linkage and the missing-opening negative case. The planned CVR mismatch fixture that can feed later comparison math is not yet implemented here.

The privacy boundary is equally important. Public observability is valuable because it lets observers check the audit path without trusting the auditor, but ballot secrecy limits what can be published. RCOUNT therefore rejects linked openings that carry candidate selections or voter identity. It also does not attempt to infer voter identity, expose secret ballot order, or turn a public opening into a vote receipt.

6 Conclusion

RCOUNT now has a narrow SOBA observable-ballot boundary. The core verifier recognizes `soba-observable-ballot-audit-v1`, requires observable ballot linkage assertions and sample steps, checks that every sampled opening exists as an inclusion proof, and rejects openings that publish choices or voter identity. The audit replay surface then reports SOBA as documented linkage rather than as completed comparison-risk replay.

This is intentionally modest. It is the public-evidence precondition for a full ballot-level comparison audit, not the full audit. The useful claim is that the observable record and opening linkage are present, well formed, and privacy bounded, so later work can attach CVR mismatch and statistical comparison checks without weakening the secrecy boundary.

References

Gio Della-Libera. Rcount overview: Reproducible election count packages. Technical report, Apportionment Research, 2026a.

Gio Della-Libera. Privacy-safe voter inclusion proofs. Technical report, Apportionment Research, 2026b.